

ALLPLAN CIVIL

MODELING OF
PRECAST GIRDER BRIDGES



CIVIL

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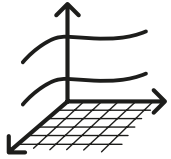
A DEDICATED WORKFLOW FOR PRECAST GIRDER BRIDGES

The geometry of precast girders is governed by the geometry of the substructure and their position along the axis. Thus, a further modeling approach is available in ALLPLAN Civil that enables users to create an exact geometry of precast girder bridges easily and quickly.

The modeling process is optimized even further by using parametric 3D templates. In this way repetitive bridge elements, such as straight precast girders, have to be defined only once and then placed parametrically as many times as necessary.

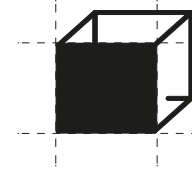
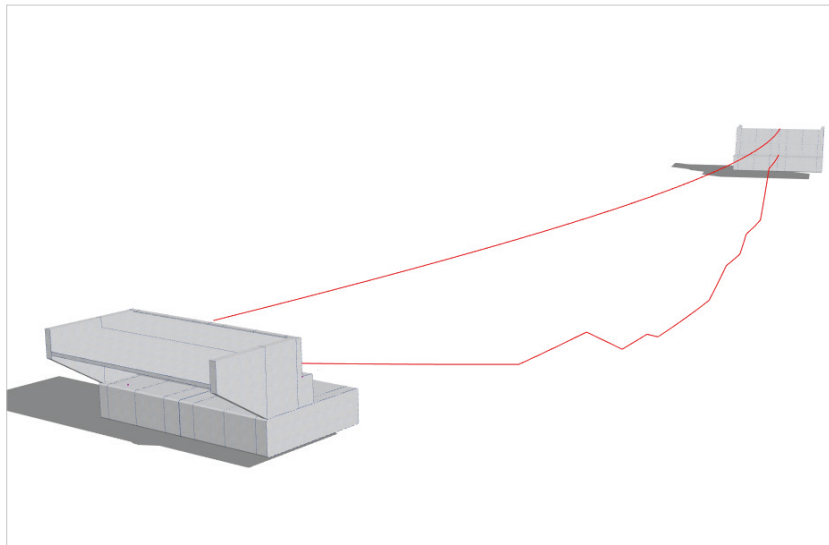
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EFFICIENT WORKFLOW WITH ALLPLAN CIVIL



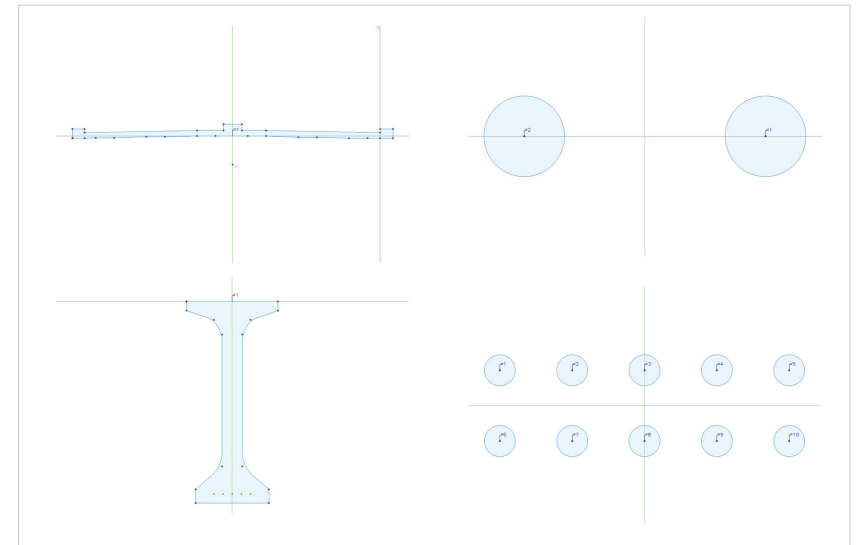
1. CREATING AXES

Every bridge construction project starts with one or more axes – with ALLPLAN Civil, you can adopt the data from an existing design (using LandXML data format) or define it manually. In both cases, the alignment is parametrically saved.

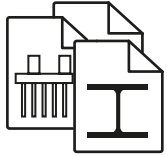


2. DEFINING CROSS-SECTIONS

You can define any cross-section and determine the geometry with its dependencies and variables. These parametric cross-sections can be adapted at any time and can be saved as a template and reused.

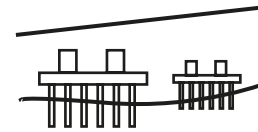
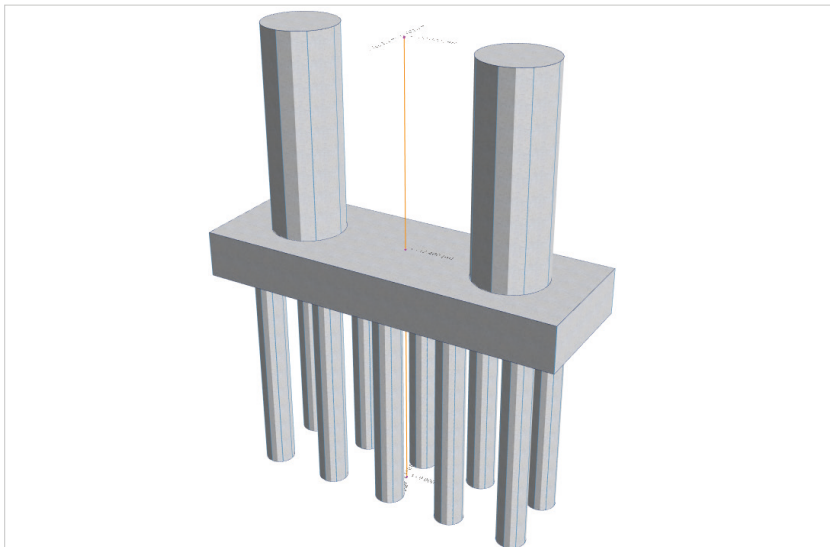


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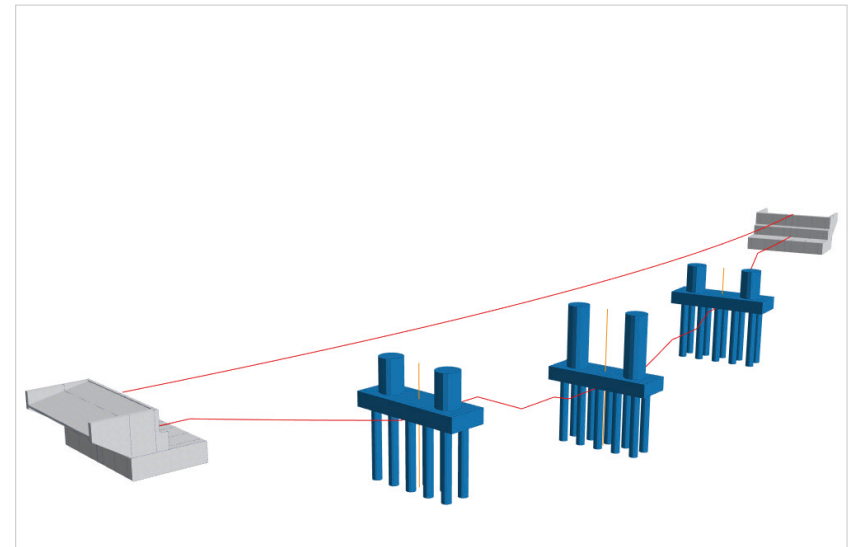
3. DESIGNING A TEMPLATE

Any geometry of a pier, foundation and precast girder can be defined as a template. In the previous step defined cross-section(s) including constant and variable parameters are used for the design of the template. If the geometry is variable, tables or formulas can be assigned as usually. Furthermore, you can set which parts of the template should be fixed and which should adjust when using it in the 3D Model.

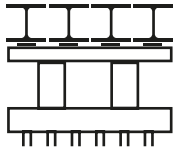


4. BUILDING A SUBSTRUCTURE

The substructure, with or without foundations, can be defined relative to one axis or relative to two axes – for example terrain axis at the bottom and road axis at the top. It can be defined directly or by usage of templates.

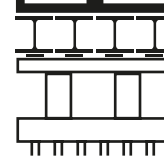
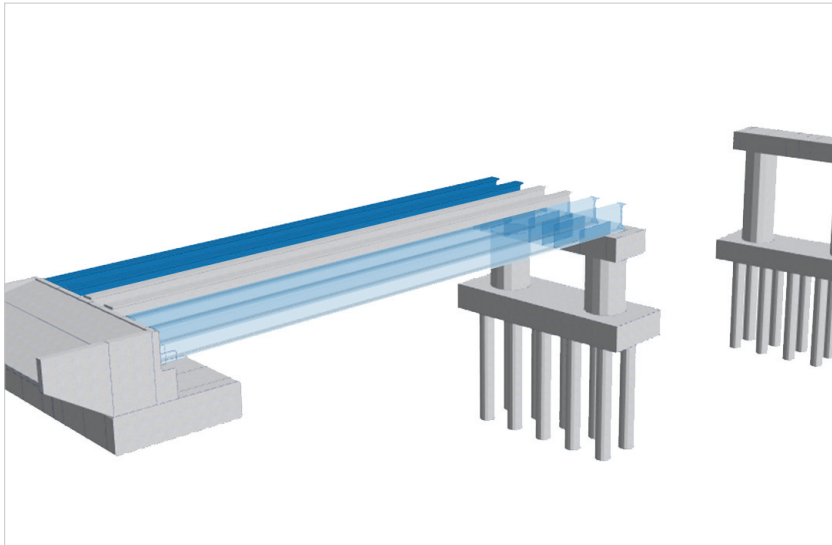


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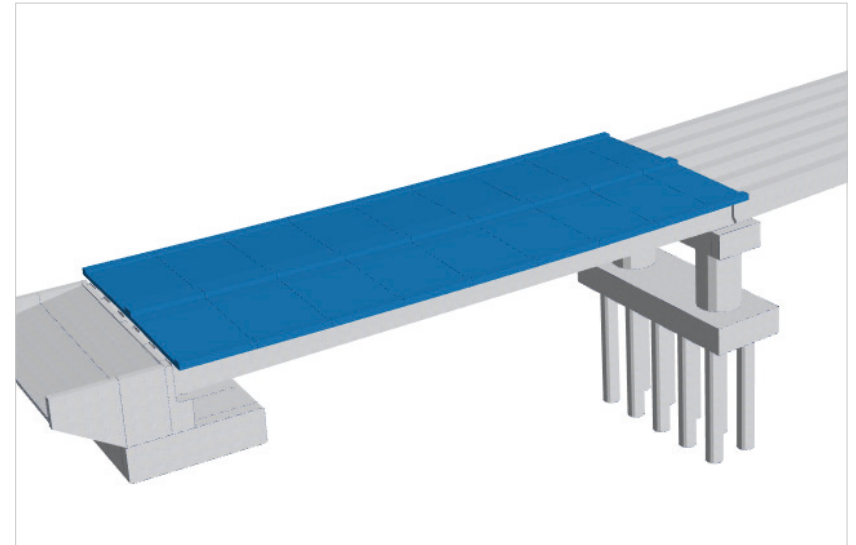
5. ASSEMBLING GIRDERS

Precast girders are, just like on the construction side, positioned on the substructure created in the previous step. Actually, they are positioned between 2 reference points (bearings). The exact position of reference points is set by the substructure geometry.

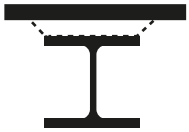


6. CONSTRUCTING A PLATE

For generating the plate geometry, the original workflow of ALLPLAN Civil is used – extruding the geometry along the axis. Also, here any variations can be used and the cross-section, and with this also the 3D model, can be equipped with all the details, either by using boundaries or by placing python parts.

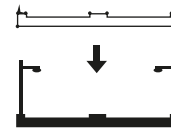
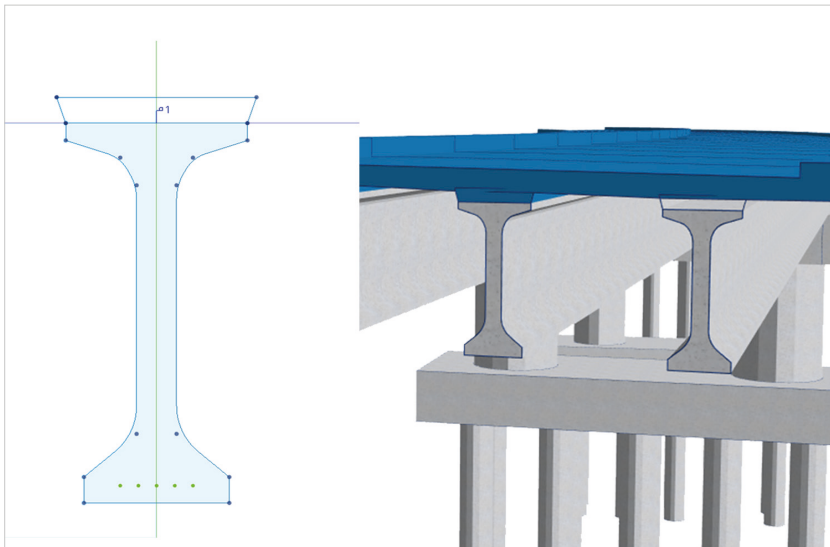


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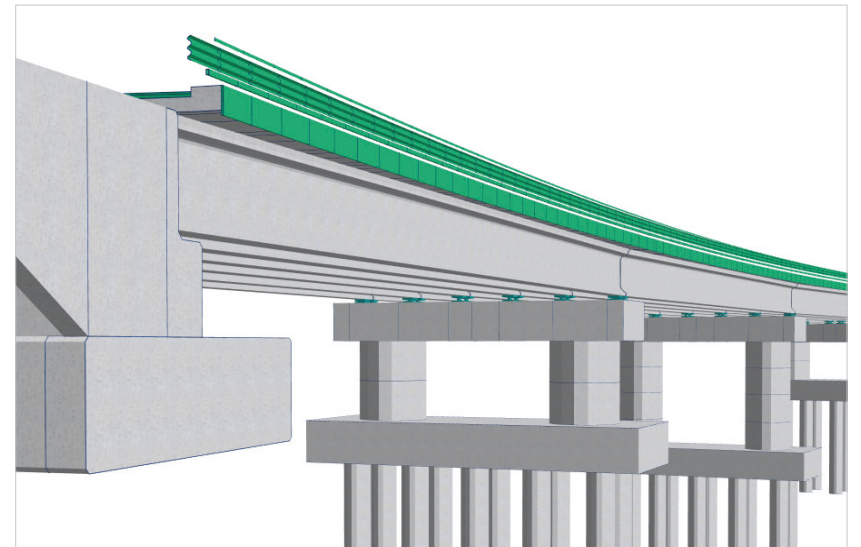
7. COMPOSING THE HAUNCH

In precast girder bridges the shape of the girders is governed by the geometry of the substructure, whereas on the other side, the geometry of the plate is governed by the axis. This is why the shape of the haunch has an arbitrary 3D geometry and varies along the girder. In ALLPLAN Civil 3D Boolean operations are used to automatically fill the space between the girder and plate, using the shape of the haunch defined in the girder cross-section.

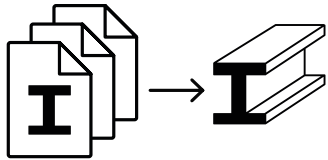


8. COMPLETING THE BRIDGE STRUCTURE

Once the main bridge elements are finished, further elements and details, like tendons, diaphragms, bearings and many more, can be generated. This can be done by modeling them or by placing parametric elements – python parts.

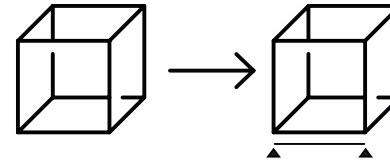
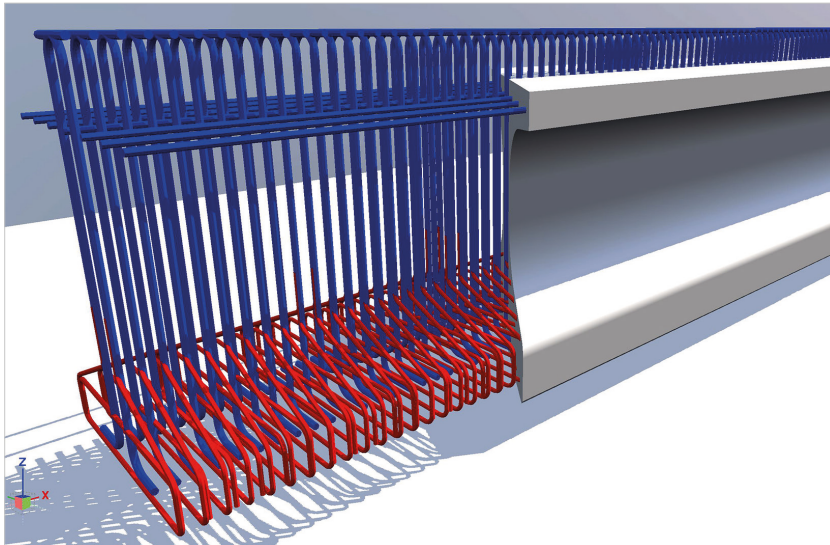


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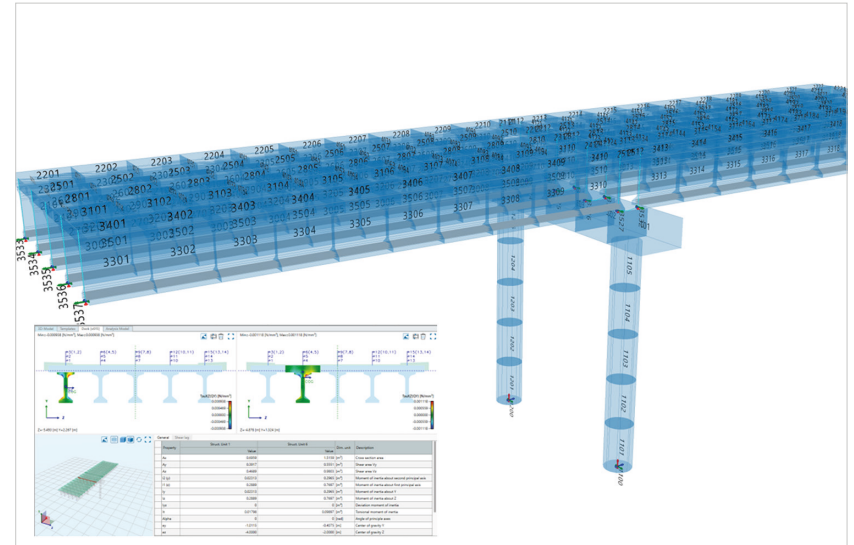
9. AUTOMATION

Parametric templates enable automation as either complete bridge models or modular components. Full-bridge templates allow key parameters to be customized for entire structures, while component templates support a flexible, element-based modeling approach. Both types include geometry, BIM data, analytical models, and reinforcement details.

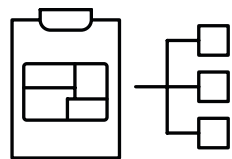


10. DERIVE STRUCTURAL ANALYSIS MODEL

Thanks to the breakthrough technology, ALLPLAN Civil semi-automatically generates the analysis model from the geometrical model. This greatly reduces the amount of work and the susceptibility to errors. Hereby the engineer retains full control by specifying structural parts and those ones which contribute as load only. One of the additional analysis-relevant definitions is the choice of generating a beam or a grillage model.

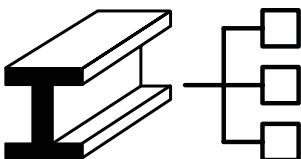
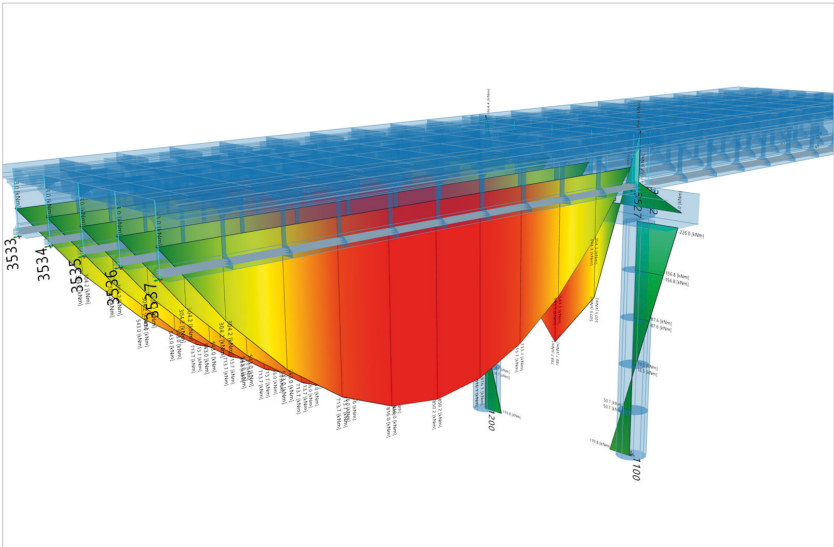


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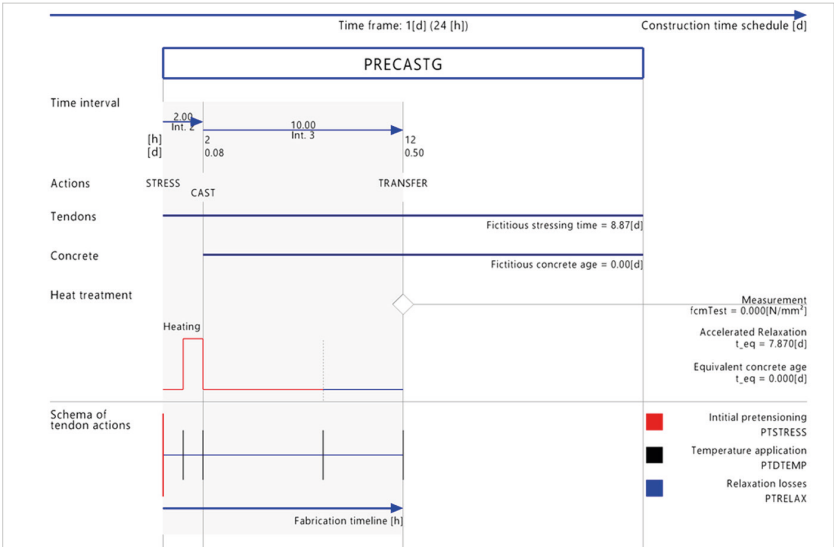
11. DEFINE CONSTRUCTION STAGES

Time as the 4th dimension is considered by easily specifying the construction process. The construction plan is divided into several phases and further into individual tasks, such as girder placement, stressing pre or post-tensioned tendons, etc. The related structural components are interactively assigned to these tasks.

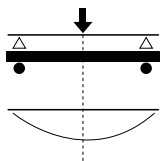


12. DETERMINE PREFABRICATION PROCES

The prefabrication process simulates key stages in the casting bed, including strand stressing, concrete casting, force transfer, and the heating cycle. This enhances the global structural analysis, particularly by incorporating adopted time-dependent effects like creep and shrinkage, and accelerated steel relaxation, all tailored to the unique behavior of precast girders during fabrication.

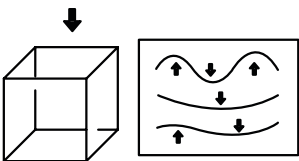
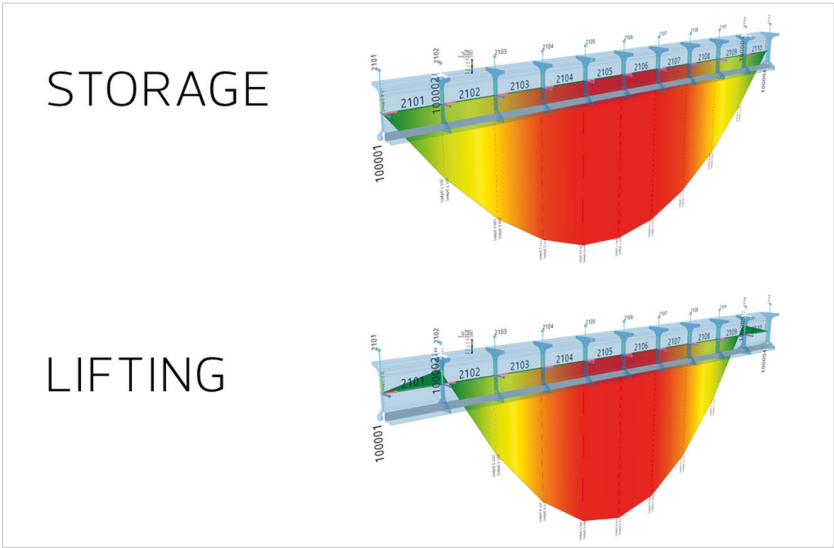


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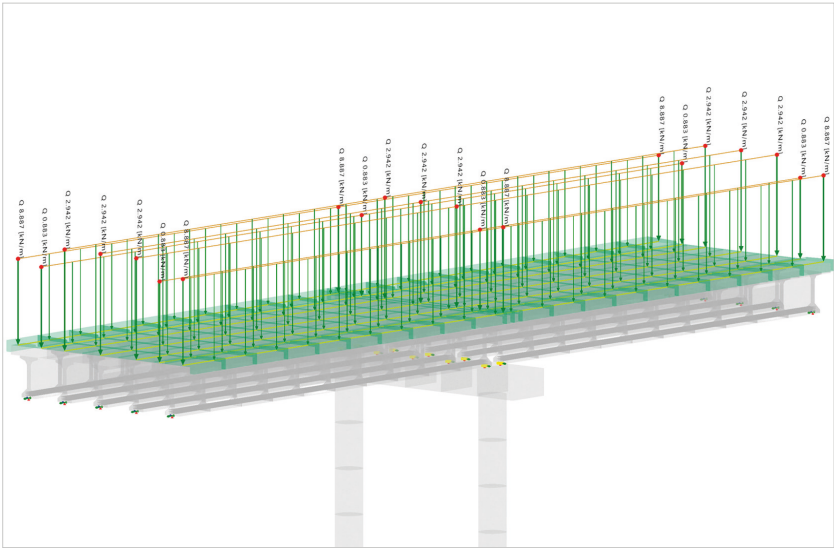
13. SIMULATE STORAGE AND TRANSPORTATION

After the fabrication process is complete and before the girder is placed in its final position, the girder undergoes multiple stages including lifting, storage, and transportation. All of this can be simulated in ALLPLAN and additionally enhances the global structural analysis and the design and sectional assessments of precast, pretensioned girders.

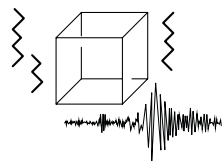


14. DEFINE ALL PERMANENT AND VARIABLE LOADS

Once the bridge is virtually constructed, the digital construction process continues with application of superimposed dead loads (e.g., sidewalk, pavement) auto-retrieved from the geometrical model. The generic approach for traffic loads allow to apply any moving load. Calculations identify the most unfavorable traffic effects, first calculating influence lines, then evaluating them with the load train and storing results in an envelope.

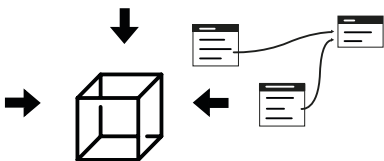


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15. EARTHQUAKE LOAD

Allplan CIVIL uses the multi-mode Response Spectrum Method for evaluating the effects of seismic loading. Mathematically, this solution is based on the excitation of the relevant natural modes and combining the different modal contributions. Internal force and displacement amplitudes related to the individual natural modes are superimposed using different methods, such as the Complete Quadratic Combination (CQC) to get the envelope of extreme values.



16. SUPERPOSITION & COMBINATIONS

The user-friendliness and usability of the superposition in ALLPLAN Civil is groundbreaking. The schematic definition of the superposition combines maximum flexibility and optimal overview. Same applies for the combinations, which are defined and visualized in a table form, giving the user an optimal overview of different combination types and load factors.

Braking-Xdr-minus

Braking

Or Braking-Xdr-plus

Or Braking-Xdr-minus

Braking-Xdr-plus

Uinf-temp-neg

Uinf-Temperature

Or Uinf-temp-pos

Or Uinf-temp-neg

Uinf-temp-pos

Temperature-unif-loading

Temperature

Or Temperature-unif-loading

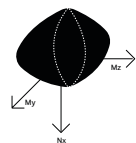
Or Temperature-grad-loading

Temperature-grad-loading

Load case/Envelope		Permanent Loads						Pre-stressing Loads				Time Effects								Name
		SW-SUM	SDL-SUM	SW-SUM	PT-SUM	PT-SUM	CS-SUM	Type 200/450	cType 310/550	cType 490/1		Add	Add	Add	Add	Add	Add	Add		
Superposition rule		Add	Add	Add	Add	Add	Add	Add	Add	Add	Add	Add	Add	Add	Add	Add	Add	Add		
Nr	Name	Type	Unfav	Favor	Unfav	Favor	Unfav	Favor	Unfav	Favor	Unfav	Favor	Unfav	Favor	Unfav	Favor	Unfav	Favor		
1	C1	Characteristic	1	1	1	1		1	1			1	1						C1	
2	C2	Frequent	1	1	1	1		1	1			1	1						C2	
3	C3	Quasi-permanent	1	1	1	1		1	1			1	1						C3	
4	C4	ULS	1.35	1	1.35	1		1	1			1	1						C4	
5	C5	ULS	1.35	1	1.35	1		1	1			1	1						C5	
6	C6	Accidental	1	1	1	1		1	1			1	1						C6	
7	C7	Basic Load - Fatigue	1	1	1	1		1	1			1	1						C7	
8	C8	Cyclic Load - Fatigue												1	1				C8	
9	C9	Cyclic Load - Fatigue													1	1			C9	
10	C10	Cyclic Load - Fatigue															1	1	C10	
11	C11	Cyclic Load - Fatigue																	C11	
12	C12	Cyclic Load - Fatigue																	C12	
13	C13	Rare	1	1	1	1		1	1			1	1						C13	

Click here to add a new row

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17. PERFORM REINFORCEMENT DESIGN & CODE CHECKING

Design of necessary reinforcement area is based on both ULS and SLS requirements. Governing combinations of internal forces including 2nd order effects are checked against flexural, torsional and shear resistance and the requirements for stress limitations and crack width. The greater of calculated or any manually specified reinforcement amount is used for code assessments of the cross-sections.

Design Code Parameters: EN 1992-2:2005

Chapter	Variable	Identification	Value	Unit
→ 24.2.3 (1)	brac		1.00	
→ 24.2.4 (1)	brac	Persistent, transient	1.50	
→ 24.2.4 (1)	brac	Accidental	1.20	
→ 24.2.4 (1)	brac	Persistent, transient	1.15	
→ 24.2.4 (1)	brac	Accidental	1.00	
→ 24.2.4 (1)	brac	Persistent, transient	1.15	

Description: Partial factor for prestressing steel

→ 24.2.4 (1)	brac	Accidental	1.00	
→ 24.2.4 (1)	brac	Persistent, transient	1.50	
→ 24.2.4 (1)	brac	Accidental	1.20	
→ 24.2.4 (1)	brac	Persistent, transient	1.15	

→ 31.2 (102)	C _{sw}		70.00	[N/mm ²]
→ 31.2 (102)	C _{sw}		30.00	[N/mm ²]
→ 31.4 (6)	β		2.50	
→ 31.6 (102)	β _{tr}		0.00	
→ 31.6 (102)	β _{tr}		1.00	
→ 32.2 (3)	f _{yk}	Lower limit of f _{yk} by 3.2.2(3)	400.00	[N/mm ²]
→ 32.2 (3)	f _{yk}	Upper limit of f _{yk} by 3.2.2(3)	600.00	[N/mm ²]
→ 32.7 (2)	ρ _{min}		0.80	
→ 33.4 (9)	ρ _{min}		1.10	
→ 33.4 (7)	ρ _{min}		0.80	
→ 5.2 (5)	β _{tr}		0.005	
→ 5.8.4 (3)	β _{tr}		1.20	
→ 5.10.2.1 (1)	β _{tr}		0.80	

Save Restore Cancel

Modify task 'CODECHECK'

General

Info

Description

Reference day

Start day

Code check

Code Check

Global Local

Final

Arguments

Day

Result name

Type of check

Add param. and inputs

Thema

45

[Seg]

Code check

Code Check

Global Local

Final

ULS combinations and groups

ULS

ULS

SLS combinations and groups

Characteristic

C1

Frequent

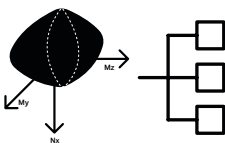
C2

Quasi-permanent

C3

OK

Cancel



18. TAILORED WORKFLOWS FOR DESIGN AND CODE ASSESMENT

Reinforcement design and code assessments can be conducted at any time, including immediately after strand release or during storage. These methods include features essential for design and assessment of pre-tensioned girders, such as accounting for accelerated concrete hardening due to heating. Concrete aging functions are adjusted using either code-specific parameters or measured concrete strength, enabling the determination of an equivalent concrete age.

25%

Girder: Deck 1 (2601 END) Stationing: 2.50000 [m]

Envelope: ULS Global Time: 36635 [days]

Ultimate Resistance in Flexure

Cross-Section

Properties

Acting Internal Forces

MAXMZ-lpt2

N [kN]

M_x [kNm]

M_y [kNm]

M_z [kNm]

Total Resultant: 4

Total Resultant: 6

Primary Effects: 4 + 5

By External Load: 4 + 5

Total: 4 + 5

Unit(4)

Unit(5)

Pretensioned tendons Y1860S7-15,2

R_{Set1}

R_{Set2}

C45/55

C45/55

A_p = 27 cm²

A_s = 59.0 cm²

B 500B

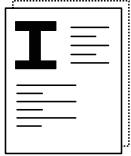
B 500B

f_{cd} (36628.92) = 25.50 MPa

f_{cd} (36585.00) = 25.50 MPa

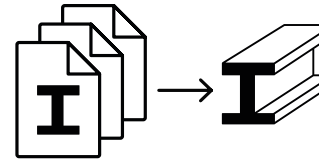
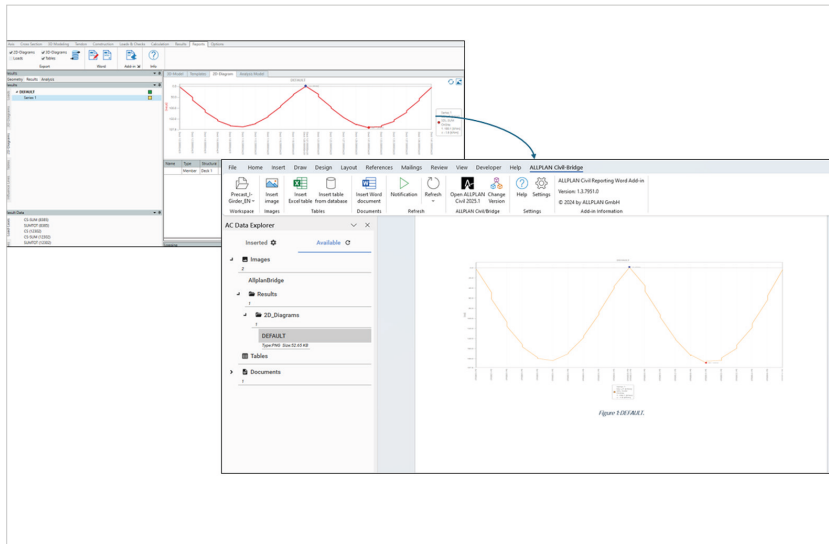
R_{axx} = 20.42 cm²/m

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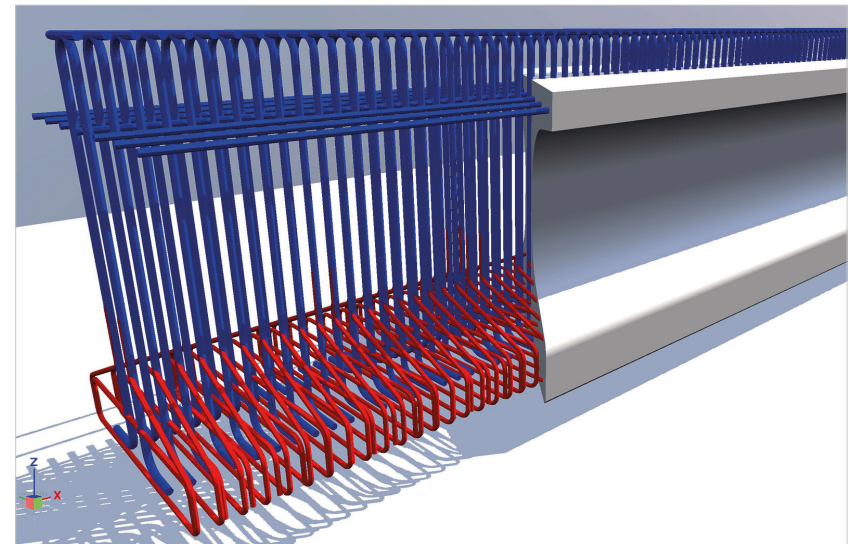
19. CREATE STRUCTURAL ANALYSIS REPORTS

The Reporting tool connects two worlds into one and with this revolutionizes the reporting process by offering a range of powerful features that enhance efficiency, accuracy, and visual impact. Bridge design and analysis data in form of tables, images, 2D and 3D diagrams, and much more, can be easily positioned and linked directly into MS Word documents. This eliminates the need for manual data entry and updates, saving time and reducing the risk of errors.

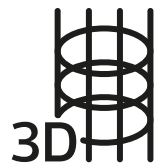


20. AUTOMATION – CONSTRUCTION TEMPLATES

Automation is implemented using predefined construction templates, specifically designed for the fabrication phase of the girder. These templates outline all key stages, including fabrication, storage, transport, lifting, and final positioning. Each construction process is governed by a unique set of variables, all of which influence the girder's behavior. By reusing these templates, manual effort in simulating the construction process is significantly reduced.

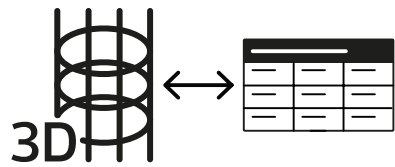
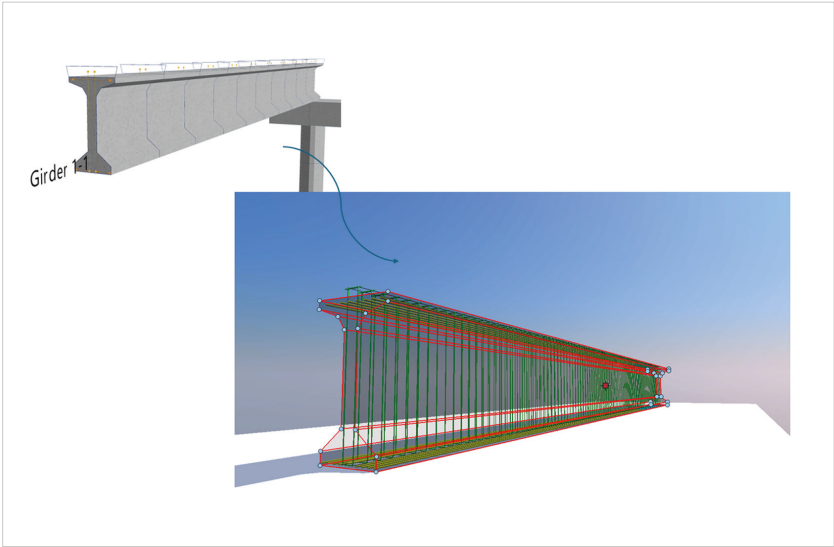


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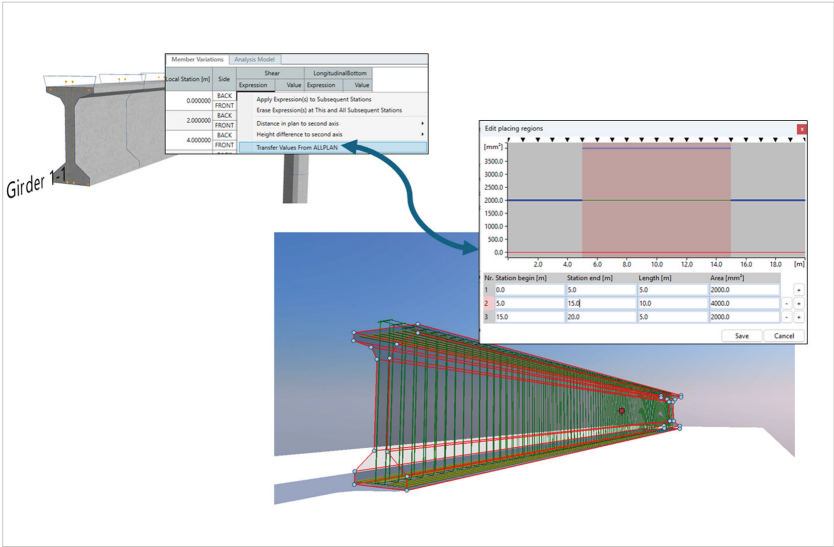
21. REINFORCEMENT DETAILING

The integrated parametric reinforcement modeling solution enhances workflows, productivity, and precision. Users can easily create parametric BIM models with reinforcement, automatically linked to ALLPLAN Python Parts technology. This integration supports reinforcement design & code checks in ALLPLAN Civil, and simultaneously automatic generation of 3D reinforcement models in ALLPLAN.

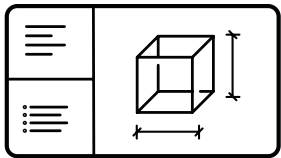


22. FINAL PROOF CHECKING

The parametric reinforcement connection is bidirectional. The reinforcement area, adjustable through stirrup spacing and the number of longitudinal bars, can be re-imported into structural analysis for final code checking.

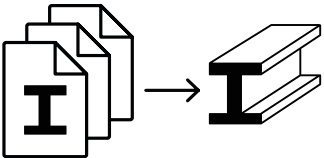
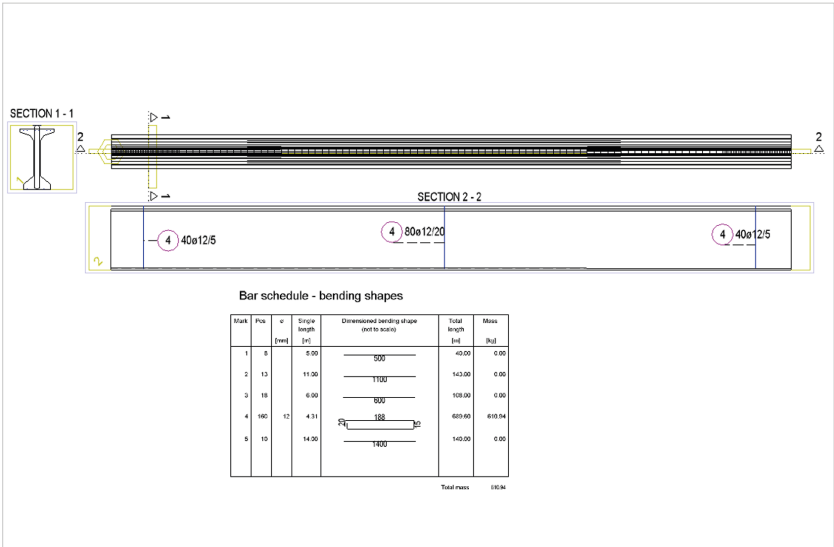


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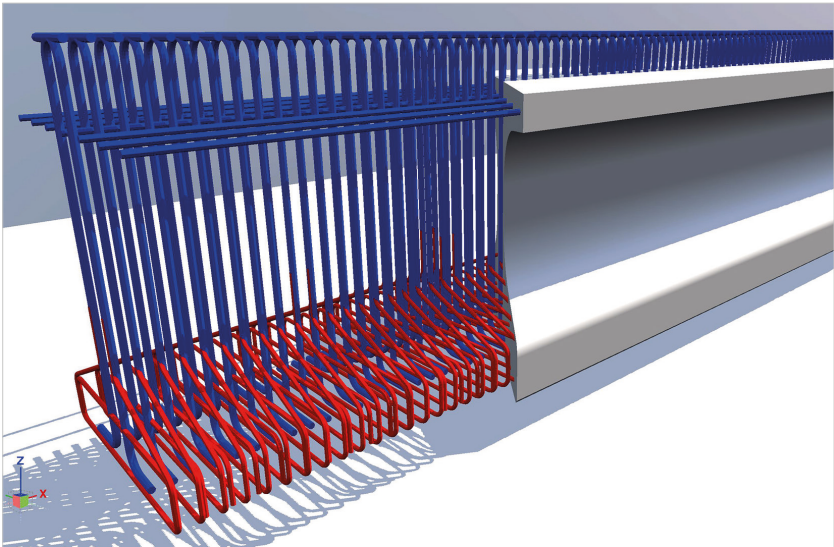
23. DRAWING GENERATION AND OTHER DELIVERABLES

Elevations, longitudinal sections along any path and transverse sections are derived from the digital 3D BIM model. CineRender from Maxon is used for realistic visualizations and ALLPLAN's powerful layout and design tools are used to create high-quality construction documentation.



24. AUTOMATION – REINFORCEMENT TEMPLATES

Once a cross section is defined and all the necessary reinforcements are set, the cross section can be used as a template across projects. The variables that control the cross section also control the reinforcement. A complete reinforcement of the structure can be created by adjusting the variables. If the complete reinforcement of a structure has to be custom created (independent of the geometry), a Python-Part template can be prepared and used as a placement at the requested positions.



About ALLPLAN

ALLPLAN is a global provider of AEC software with BIM solutions for architecture, structural engineering, detailing, fabrication and construction. True to our "design to build" claim, we provide tools that enable earlier data-driven design decisions, support digital fabrication and leverage information throughout the entire construction process. Integrated cloud technology further optimizes interdisciplinary collaboration on building and infrastructure projects. Our innovative workflows empower architects, engineers, and construction professionals to deliver their projects more productively, safely, and eco-consciously.

Around the world, over 700 dedicated employees continue to write the ALLPLAN success story. Headquartered in Munich, Germany, ALLPLAN is part of the Nemetschek Group – a pioneer for digital transformation in the construction sector.

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